Preetham Bhat

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LeetCode | Portfolio

Summary

AI Engineer and Backend Developer focused on building intelligent, reliable, and scalable software solutions. Driven by solving real-world problems through innovative applications and efficient software engineering.

Education

Bachelor of Engineering, Vidyavardhaka College of Engineering *2022-2026*
Information Science Engineering **CGPA: 8.53**
Coursework: Data Structures and Algorithms, Computer Networks, Operating Systems, RDBMS, Software Engineering, Design and Analysis of Algorithms.

Work Experience

Internship Program — IBM SkillBuild **Jul 2024 – Aug 2024**
Remote - Mysore, India

- Engineered an advanced **Dialogflow** chatbot integrated with REST APIs, enabling **24/7 automated support** and reducing repetitive query handling by leveraging **HTML/CSS, JavaScript, and Python**.
- Optimized intent classification pipelines, boosting accuracy and reducing response latency through refined datasets and in-depth analytics using **Pandas, NumPy, and Scikit-learn**.

Projects

Multi-Plant Disease Detection **Tech Stack: Python, FastAPI, React, TensorFlow, YOLOv8, Firebase**

- Developed an **AI-powered plant disease detection system** supporting **3 crops**, trained on **8,158 images** with **12,913 annotated objects** across **6 classes**, using **YOLOv8 and TensorFlow**.
- Designed an end-to-end **deep learning inference pipeline** integrating object detection, disease classification, severity assessment, and treatment recommendations through **FastAPI REST APIs** and a **React** frontend.
- Implemented **Firebase Authentication** and modularized training, inference, and application components to improve code maintainability, model integration, and deployment readiness.

AI-Driven Waste Management **Tech Stack: React, Tailwind, FastAPI, Python, MobileNetV2, BERT**

- Built an AI-powered waste classification platform achieving **98% accuracy** on **25K+ images** using **MobileNetV2**, and integrated voice-command functionality with **BERT** for improved accessibility and interaction.
- Increased engagement and usability by introducing gamification and providing real-time, location-based disposal suggestions through optimized **FastAPI** endpoints, enhancing API efficiency.

Match Alert Microservices **Tech Stack: Node.js, Express.js, MongoDB, RabbitMQ, Docker, Jest**

- Created a **microservices-based backend** for esports match alerts using **RabbitMQ** for message queueing. Enabled scalable, asynchronous processing to handle high-frequency alert events with low latency. Deployed **Dockerized microservices**, improving delivery reliability and fault tolerance.

Technical Skills

- **Programming Languages:** Python, Java.
- **Backend Frameworks:** FastAPI, Node.js, Express.js, REST APIs.
- **AI/ML Frameworks:** TensorFlow, PyTorch, Scikit-learn, YOLOv8, BERT, Pandas, NumPy.
- **Databases & Caching:** MySQL, MongoDB, Redis.
- **Tools & DevOps:** Git, GitHub, Docker, Postman, GitHub Actions, CI/CD, Render.
- **Other Technologies:** React.js, JWT, OAuth 2.0, RabbitMQ, Jest.

Achievements

- Secured **4th place** at **Infothon 4.0**, a national-level hackathon, by developing an **AI-driven waste management system** achieving **98% accuracy**.
- Led a cross-functional team to **5th place** in a national-level **coding and UI/UX design competition**.

Certifications

- Developing Backend Applications with Node.js and Express.js – Meta (Feb 2024)
- Advanced React and Principles of UI/UX Design – Meta (Feb 2024)
- Google Cloud Career Launchpad – Cloud Engineer – Google (Dec 2024)